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Wide area illumination Raman scheme for simple and nondestructive discrimination of seawater cultured pearls

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Raman spectroscopy, along with discriminant partial least squares (PLS), was successfully used to discriminate among three different groups of cultured pearls (fresh water, Akoya and South seawater). The discrimination between Akoya and South seawater pearls using XRF (X-ray fluorescence), one of the most frequently adopted analytical methods in pearl analysis, has been especially difficult owing to their similar mineral compositions. The selective Raman features helped in effectively discriminating between these two pearl groups. The difference in the intensities of the $CaCO_3$ bands of Akoya and South seawater pearls provided a valuable clue. Along with the selective Raman feature, a reproducible Raman spectral collection achieved using a wide area illumination (WAI) scheme played an important role in the determination of the pearl groups, although the pearls were hard-surfaced, round, solid samples of different sizes and surface shapes. Unwanted spectral variation originating from sensitivity to sample placement relative to the focal plane and from unsuccessful sample representation due to the probing of a localized area, factors that could possibly deteriorate Raman reproducibility, were substantially lessened using the WAI scheme. ATR (attenuated total reflection) IR spectroscopy requiring direct contact with the pearl could be inadequate for discrimination or classification where large numbers of repeating and reproducible measurements are required. **Copyright © 2009 John Wiley & Sons, Ltd.**

Keywords: pearl discrimination; aragonite; wide area illumination Raman scheme; discriminant partial least squares

Introduction

The demand of objective and fast identification of diverse cultured pearls for fair evaluation is growing because domestic and international trade of pearls has greatly increased. Generally, three kinds of cultured pearls are most frequently encountered in market: fresh water, Akoya and South seawater, although diverse kinds of pearls are available depending on mollusks for cultivation such as *Pinctada fucata, Pinctada maxima, Pinctada margaritifera, Hyriopsis schlegelii* and so on. Both Akoya and South seawater pearls are cultured in seawater. A pearl is approximately composed of 93–95% $CaCO_3$ (as aragonite), 5% organic matrix (including hard proteins, β-chitin and polysaccharides) and 1% moisture.[1–4] The outer part of a pearl is composed of alternating sheets of $CaCO_3$ and organic matrix. Figure 1 shows actual pictures of the three kinds of pearls in different sizes. The sizes of the fresh water and Akoya pearls are similar, while the sizes of the South seawater pearls are larger. Recently, smaller South seawater pearls similar in size to Akoya pearls have been cultured and are available on the market; therefore, a methodology to correctly discriminate between Akoya and South seawater pearls is now in demand. Additionally, it should be nondestructive as destructive sampling largely degrades the value of pearls.

The most frequently adopted analytical method for the identification of pearls is X-ray fluorescence (XRF), which can determine the mineral content of the pearls.[5] XRF is very effective in discriminating between fresh water and seawater cultured pearls as the content of manganese (Mn) is higher in fresh water pearls, and the content of strontium (Sr) is higher in seawater cultured pearls. However, the compositional analysis of minerals cannot allow one to differentiate between two seawater cultured pearls (Akoya and South seawater) because their mineral compositions are very similar to each other.[5] Consequently, an analytical method other than X-ray-based analysis capable of discriminating between Akoya and South seawater pearls is highly needed.

Raman spectroscopy could be a strong candidate for the discrimination between Akoya and South seawater pearls as it provides rich organic and inorganic chemical information. In pearl analysis, Raman spectroscopy has been mainly focused on the structural study of nacre including polymorphism of biogenic carbonate and the corresponding vibrational band assignment.[6–8] Additionally, it was also used for the preliminarily investigation of the difference between natural and artificially colored cultured pearls, and the relevant bands assignments.[9] The origin of coloration, crystalline formation and pigments has been extensively studied using Raman microscope with diverse excitation wavelengths.[10–12] However, the discrimination between Akoya and South seawater pearls using Raman spectroscopy in conjunction with chemometric analysis that can impose statistical significance of the discrimination result has not been studied so far.

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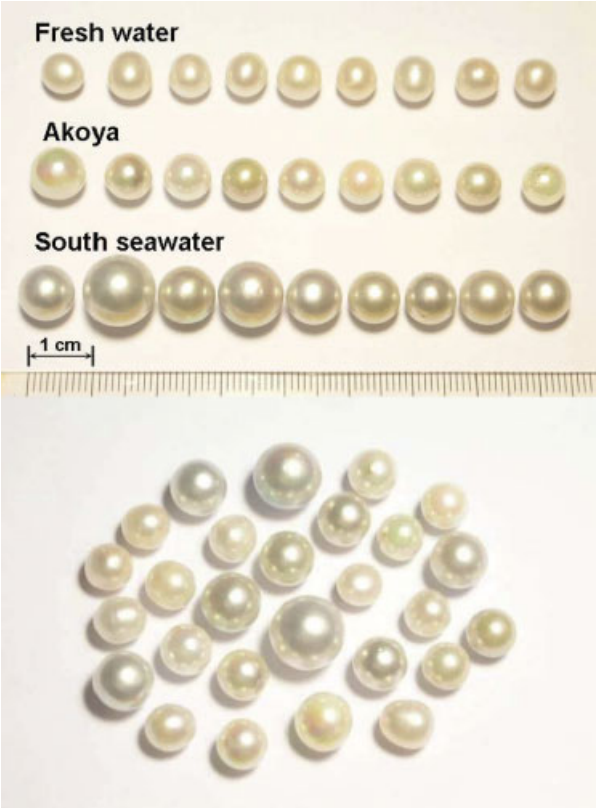

Figure 1. Nine pearls from each origin (top), and the same pearls randomly mixed (bottom).

In this publication, we have attempted to differentiate two seawater cultured pearls (Akoya and South seawater) using Raman spectroscopy incorporating a unique spectral collection strategy, the so-called the wide area illumination (WAI) scheme. The prerequisite for reliable discrimination or classification is the acquisition of reproducible Raman spectra, especially a sample that is round with varying shapes and morphologies such as pearls (refer to Fig. 1). The resulting reproducibility of Raman measurements is degraded if they are sensitive to the placement of the sample relative to the focal plane, or if they fail to correctly represent the sample because only a small area is probed. The WAI scheme applies the laser beam to a wide area of the sample for Raman spectral collection (6 mm laser spot with a focal length of 248 mm).[13] Any unwanted spectral variation originating from sample placement or from varying surface shape can be greatly reduced because of the long focal length. Additionally, by covering a large sample area, the sample represented by the Raman measurement is more reliable. Using the collected Raman spectra of the pearls using the WAI scheme, discriminant partial least squares (PLS) were applied to discriminate Akoya and South seawater pearls. Additionally, the discrimination between fresh water and seawater cultured pearls has been also investigated using the same analytical approach, although it can also be feasible with the use of X-ray-based analysis.

Experimental

Samples

Some pearls can be easily identified by simple visual examination of traits such as color, but two different types of pearls can have a very similar visual appearance. In this study, we have tried to exclude pearls that can be easily distinguished by sensory examination and have collected pearls with more visual similarity. Fourteen fresh water pearls, nine Akoya pearls, and twenty South seawater pearls were acquired from the Korea Pearl Laboratory, Seoul, Korea. Fresh water cultured pearls were obtained from *Hyriopsis cumingii* shells cultured at China. Akoya and South seawater pearls were obtained from *Pinctada Fucada* (cultured at Japan) and *Pinctada Maxima* shells (cultured at Australia), respectively. Nine pearls from each origin are shown at the top of Fig. 1. The average diameters of the fresh water, Akoya and South seawater pearls are approximately 6.0, 6.5 and 7.8 mm, respectively. As shown, some South seawater pearls are larger than the pearls in the other two groups, while some of them are of similar size. The color of the South seawater pearls looks darker in the picture, but they are actually fairly white. The bottom picture shows the same samples randomly mixed. Except for five or six large pearls, it is not easy to distinguish the groups of pearls by visual examination without experience and education.

Raman spectral collection

The schematic diagram of the WAI scheme is explained in detail in a previous publication.[13] The WAI scheme used in this study was developed by Kaiser Optical Systems (Ann Arbor, MI, USA). An excitation laser (785 nm, diode laser) was magnified to form a circular illumination area with a diameter of 6 mm (area: 28.3 mm^2) to cover a large sample area. The scattered radiation was then collected by an array of 50 optical fibers and delivered to the CCD detector. The pearl samples were simply positioned at the focal point of the laser for the spectral collection. Each Raman spectrum represents an accumulation of eight scans, with an exposure time of 5 s for each scan (acquisition time: 40 s, spectral resolution: 4 cm^{-1}). Triplicate spectra were collected for each sample. During triplicate spectral collection, a pearl sample was rotated and repositioned, thereby ensuring that the sample orientation would be random. Discriminant PLS as well as principal component analysis (PCA) were performed using Matlab Version 7.0 (MathWorks Inc., MA, USA).

Results and Discussion

Spectral features and principal component analysis

Figure 2 shows the normalized Raman spectra of three classes of pearls (fresh water, Akoya and South seawater) by averaging all spectra for a given class. The baselines of the raw Raman spectra were corrected at eight different wavenumbers (1850, 1400, 1189, 973, 807, 685, 365 and 90 cm^{-1}) and the peak area under the 1850–90 cm^{-1} range was calculated. Then, each baseline-corrected spectrum was divided by the corresponding peak area under that range. Through the normalization, we reduced the possible Raman peak intensity variation resulting from a slight change in the sample position with respect to the focal point and variation in the surface shape. The top left plot in the figure shows the major Raman bands of the pearls (located at 1462, 1086, 704, 208 and 154 cm^{-1}) in the entire spectral range (1800–100 cm^{-1}). For a more detailed examination, the spectral range was divided into eight narrower ranges to highlight the low-intensity Raman bands that could also provide valuable clues to distinguish among the three different pearl groups. In each plot of the narrow spectral range, the *y*-scale

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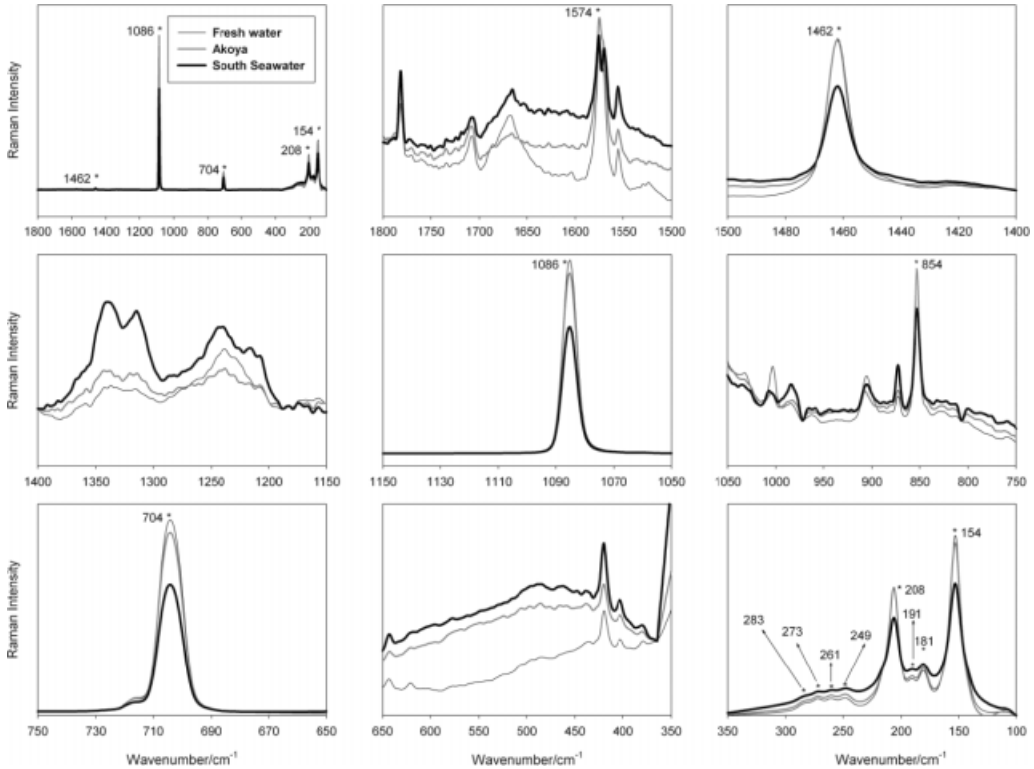

Figure 2. Average normalized Raman spectra of three classes of pearls used in this study (fresh water, Akoya and South seawater). The spectral features are present in the whole spectral range (top left plot), as well as in eight narrow spectral ranges. The peaks marked by asterisks correspond to vibrations from aragonite ($CaCO_3$), the major constituent of the outer shell of a pearl.

was adjusted to maximize the spectral comparison. The peaks marked by asterisks correspond to vibrations from aragonite ($CaCO_3$), the major constituent of the outer shells of pearls. The positions of these bands match with those found in a previous publication.[6,7,14,15]

On the basis of the examination of the four highest intensity bands (1086, 704, 208 and 154 cm^{-1}, designated in the top left plot of the whole spectral range) corresponding to $CaCO_3$, the order of Raman intensities is analogous. These four $CaCO_3$ bands in the fresh water and South seawater pearls show the highest and lowest intensities, respectively, while the intensities of the fresh water and Akoya pearls are more similar as shown in Fig. 2 (the 1800–100 cm^{-1} range). This observation indicates that the crystallinity of $CaCO_3$ varies across the three groups of pearls. As generally known, the intensity of a Raman band is sensitively varying to crystalline structure. The different intensities of $CaCO_3$ in a pearl, as shown in Fig. 2, should provide one of the most important pieces of spectral information, especially for the discrimination between two seawater cultured pearls (Akoya and South seawater). It is reasonable to expect that the crystalline formation of $CaCO_3$ in Akoya and South seawater pearls is quite different from each other, although a further in-depth study will be required to elucidate the difference based on crystallography. The 350–100 cm^{-1} range, where the Raman bands of $CaCO_3$ are only present, also shows discernable spectral differences between the Akoya and the South seawater pearls. The band intensities at 208 and 154 cm^{-1} differ significantly.

As shown in Fig. 2, the use of the $CaCO_3$ bands would not be efficient for discriminating between the fresh water and seawater cultured pearls as the intensities of these bands in fresh water and Akoya pearls are similar. The use of additional selective Raman bands would be required to improve the differentiation. Minute spectral differences between fresh and seawater cultured pearls were observed in the 1800–1500 cm^{-1} range. The 1520-cm^{-1} band for fresh water pearls was relatively high, while there was no specific feature around that at 1520 cm^{-1} for the two seawater cultured pearls. On the basis of the report by Karampelas *et al.*[10], the 1520 cm^{-1} band corresponds to the vibration of C=C present in polyene-based natural pigments. A recent study shows that a small amount of pigment components is also present in cream/white pearls and the corresponding weak band at 1520 cm^{-1} is observed.[11] Additionally, the 1005 cm^{-1} band present in the 1050–750 cm^{-1} range provides the selective feature to allow us to discriminate between fresh and seawater cultured pearls. It is difficult to determine the origin of this band at this stage; however, it is presumed that the 1005 cm^{-1} band corresponds to polyene.[16,17] Nonetheless, the observed spectral difference can be effectively used for the discrimination. Overall, the spectral differences present in the 1800–1500 and 1050–750 cm^{-1} ranges can be used to differentiate between fresh water and seawater cultured pearls.

Discriminant partial least squares

To differentiate the pearls into two groups, discriminant PLS was used.[18,19] This method belongs to the category of supervised learning, where prior knowledge of the class membership is used for discrimination. With knowledge of the existence of two groups, an arbitrary number is assigned to each group and then PLS is performed using the arbitrarily assigned numbers as reference data. In this study, we assigned one group as 25 and the other as 75. When the value of a sample predicted using PLS was under 50,

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Table 1. Discrimination errors for the differentiation between Akoya and South seawater pearls, as well as fresh water and seawater pearls. The whole spectral range (top) and eight different spectral ranges were used for discriminant PLS. The numbers in parentheses correspond to the numbers of factors used for PLS

Range (cm^{-1})	Akoya/South seawater (%)	Fresh water/Sea water (%)
1800–100	0.0(3)	24.7(2)
1800–1500	2.3(3)	0.8(4)
1500–1400	4.6(3)	3.9(3)
1400–1150	9.3(6)	3.2(4)
1150–1050	12.3(2)	24.7(2)
1050–750	3.0(4)	0.0(6)
750–650	7.7(1)	23.3(1)
650–350	9.2(2)	16.1(2)
350–100	0.0(3)	24.0(3)

it was classified into the first group. The sample was classified into the second group if the value was over 50. As mentioned earlier, we initially differentiated the Akoya and South seawater pearls using discriminant PLS and then the same strategy was applied to differentiate the fresh water and seawater cultured pearls.

As the number of samples used in this study was not large, we cross validated each dataset by dividing it into four segments to improve the statistical significance of the resulting classification errors. For every given range, a PLS model was built using three segments (75% for the calibration and 25% for the validation). The one remaining segment was predicted, and the corresponding discrimination error was calculated as a percentage. This procedure was repeated four times until all of the samples were predicted. Finally, the average discrimination error was obtained.

It would be worthwhile to evaluate the whole spectral range as well as all narrow spectral ranges for the discrimination, as the necessary spectral differences could be present throughout the spectral range or localized in a narrow spectral range. The whole spectral range and the eight narrow spectral ranges shown in Fig. 2 were used for discriminant PLS, and the corresponding discrimination errors were compared. Table 1 shows the resulting discrimination errors obtained using the whole spectral range and the eight narrow spectral ranges. The numbers in parentheses correspond to the numbers of factors used for PLS. We only used the factors describing spectral variation of at least 1% out of the total variation present in a given range.

The major $CaCO_3$ bands could be used to easily discriminate between two seawater cultured pearls (Akoya and South seawater) because the Raman intensities of these bands differ significantly. In other words, the use of the full spectral range is an appropriate choice as the spectral features of $CaCO_3$ are widely spread across the entire range. As expected, the best result was achieved using the entire spectral range. The use of the range from 350 to 100 cm^{-1}, where only the $CaCO_3$ feature was present with a systematic spectral variation to follow the difference, also provided perfect discrimination. Figure 3(a) shows the score scatter plot created using the first and second PLS scores, which were generated using the 350–100 cm^{-1} range. Two groups of Akoya and South seawater pearls are clearly identified.

For the discrimination between fresh water and seawater cultured pearls, the use of the 1050–750 cm^{-1} range led to perfect classification. The second best result was achieved using the spectral information in the 1800–1500 cm^{-1} range. As described in Fig. 2, these two ranges contained more selective bands to use

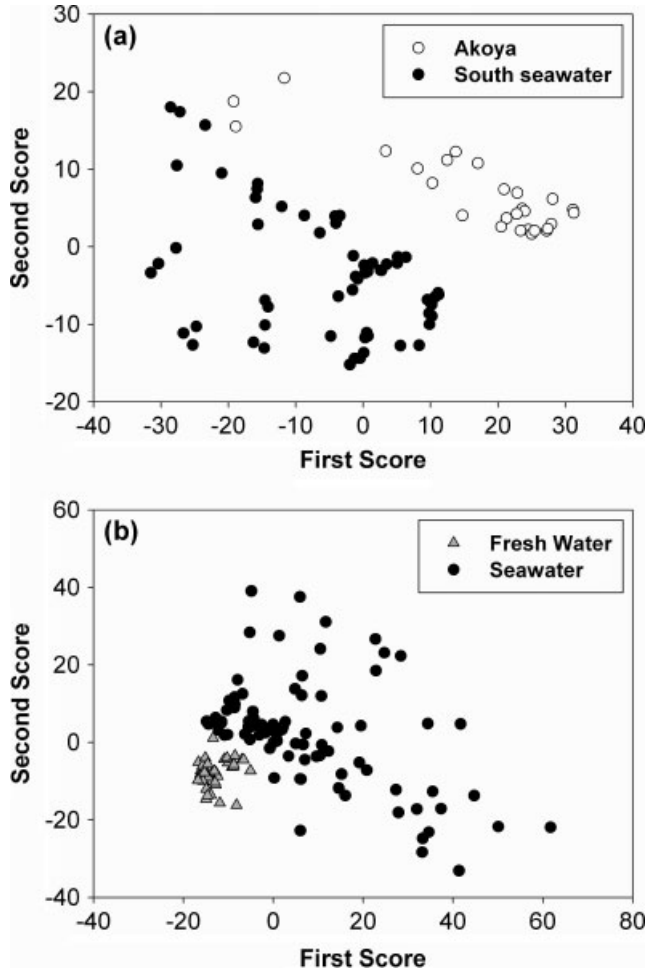

Figure 3. Score scatter plot showing the discrimination between Akoya and South seawater pearls using the 350–100 cm^{-1} range (a), and the discrimination between fresh water and seawater cultured pearls using the 1050–750 cm^{-1} range (b).

for discrimination; therefore, the results were significantly better than those from other spectral ranges. Figure 3(b) shows the score scatter plot created using the first and second PLS scores, which were generated using the 1050–750 cm^{-1} range. Two isolated groups of fresh water and seawater cultured pearls are clearly observed.

Reproducibility test

The selective Raman features in the spectra from the three different pearl-types led to the successful discrimination performance as shown in Table 1. However, the Raman spectral feature would have little use if the corresponding Raman measurement was not reproducible. When the magnitude of spectral variation originating from an irreproducible measurement was larger than that of the spectral variation necessary to identify the groups of the pearls, the resulting discrimination performance would be substantially degraded. To evaluate the reproducibility of our method, we collected 20 spectra from one fresh water pearl over a 2-day period. For each measurement, the sample was repositioned, thereby ensuring that the sample orientation and position for each measurement was random.

PCA[20] was performed by combining all of the spectra in the dataset with the 20 newly collected spectra from the single

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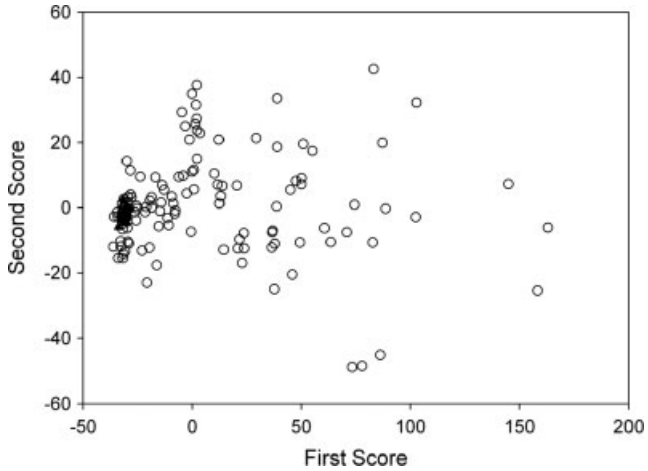

Figure 4. Score scatter plot including the scores from the whole dataset and the scores from 20 repeated measurements of the same sample. The whole spectral range of 1800–100 cm^{-1} was used for PCA. Open circles and filled triangles correspond to the whole dataset and the 20 repeated measurements respectively.

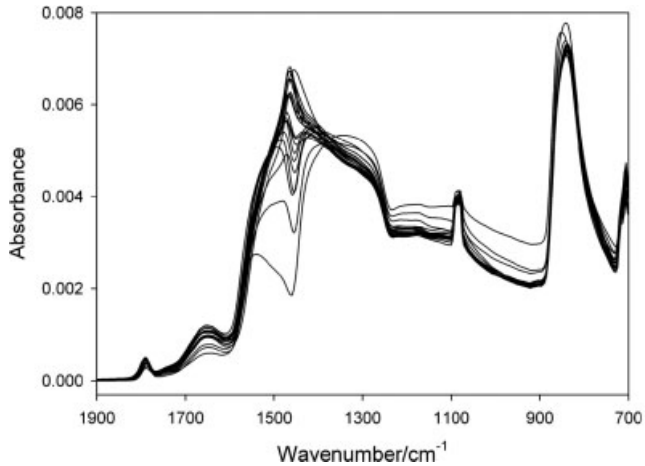

Figure 5. Twenty ATR-IR spectra collected for the same pearl. The baselines of the spectra were corrected at 1900 cm^{-1}.

pearl, and the resulting score distribution was evaluated. The full spectral range of 1800–100 cm^{-1} was used for PCA. By comparing the degree of score scatter of the data set with that of the 20 repeated measurements, the reproducibility of Raman collection using the WAI scheme can be evaluated. The resulting score scatter plot, generated from the first and second scores, is shown in Fig. 4. Basically, the scores of the 20 spectra are presented on the top of the scores from the whole dataset. For easier comparison, open circles have been used to designate the whole dataset, and filled triangles have been used for the 20 repeated measurements. Compared to the distribution of scores in the dataset, the scores of the repeated measurements aggregate tightly together without any severe deviation from the cluster, indicating acceptable reproducibility of Raman collection using the WAI. If the conventional Raman scheme that illuminates small areas is used for the same purpose, the resulting reproducibility is degraded due to the variation in the Raman peak resulting from changes in the sample position relative to the focal point as well as from variations in the surface shape, as can be extrapolated from previous publications.[21,22] Additionally, the possibility of insufficient sample representation that results from probing localized chemical information is eliminated when using the WAI scheme.

Along with Raman spectroscopy, IR spectroscopy can also provide valuable information relating to the composition and structure of a pearl. The most challenging practical hurdle for the analysis of pearls using IR spectroscopy is the difficulty in collecting reproducible IR spectra. As the sample is a sphere with a hard surface and destructive sampling is not recommended, ATR (attenuated total reflection) would be a proper way to acquire IR spectra. To collect the IR spectrum of a pearl, an ATR element is required for firm and reproducible contact with the sample. As is generally known, the reproducibility of an ATR-IR measurement can be influenced by diverse factors such as contact area, contact pressure and surface morphology. We collected ATR-IR spectra from the same fresh water pearl using an FT-IR spectrometer equipped with a microscope and a diamond ATR element (SensIR Technologies, Danbury, CT, USA). The aperture size was 100 μm in diameter and an MCT detector was used. During IR spectral collection, we focused on maintaining constant contact with the ATR element during 20 independent spectral collections.

Figure 5 shows the resulting 20 IR spectra in the range from 1900 to 700 cm^{-1}. The baselines of the spectra were corrected at 1900 cm^{-1}. The band shapes varied greatly, even though they were collected from the same sample. Although there are several major bands that can be used to determine the chemical identity of a pearl, they would be difficult to use for discrimination because of their degraded reproducibility. It was difficult to maintain constant contact between the ATR element and the pearl during each measurement. Additionally, we have tried collected IR spectra using an IR microscope in the reflection mode; however, reasonable spectral features were hardly obtained.

Conclusion

The WAI scheme was very effective in collecting reproducible Raman spectra from round, solid, hard-surfaced samples of varying sizes and surface shapes such as pearls. The spectral collection was easy, requiring only that a pearl be positioned at the focal point. By using selective Raman bands, the discrimination between Akoya and South seawater pearls was possible using discriminant PLS, a task that is difficult when using XRF. Additionally, a discrimination was also demonstrated between fresh and seawater cultured pearls. Recently, different mollusks have been used for cultivation, and cultured regions become more diverse; therefore, the sample set used in this study might not be enough to represent all possible variations and a discrimination error might marginally increase with a larger sample-size. However, the proposed Raman scheme has a strong potential for simple, fast and nondestructive evaluation of cultured pearls along with other existing analytical methods.

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